



Parul University
Faculty of Engineering and Technology
Parul Institute of Engineering and Technology
Computer Science & Engineering Department

Subject Name	Data Analytics and Data visualization	A.Y	2026-27
Subject Code	303105314	Semester	5th
Sr No	Question	COs	B.T
1	Explain the different types of Data Analytics with examples.	CO1	L2 – Understand
2	Compare Data Analysis and Data Analytics.	CO1	L4 – Analyze
3	Discuss the role of a Data Scientist and a Data Analyst.	CO1	L2 – Understand
4	Describe the classification and elements of Data Analytics.	CO1	L2 – Understand
5	Why is Data Analytics important in today's industry?	CO1	L4 – Analyze
6	Explain the importance of Python in Data Analytics.	CO2	L2 – Understand
7	Differentiate between Population and Sample in statistical analysis.	CO2	L4 – Analyze
8	Describe the different levels of data measurement.	CO2	L2 – Understand
9	Discuss measures of central tendency and dispersion with examples.	CO2	L3 – Apply
10	What is a confidence interval? Explain with an example.	CO2	L3 – Apply
11	Define and explain different types of probability distributions.	CO3	L2 – Understand
12	What is hypothesis testing? Describe its steps.	CO3	L2 – Understand
13	Explain the use of Chi-Square test with an example.	CO3	L3 – Apply
14	Discuss Sampling and Sampling Distribution in Data Analytics.	CO3	L2 – Understand
15	What is ANOVA test and when is it used?	CO3	L3 – Apply
16	Differentiate between Linear and Logistic Regression.	CO4	L4 – Analyze
17	Explain the K-Means Clustering Algorithm with an example.	CO4	L3 – Apply
18	Discuss Decision Tree Classification and its working.	CO4	L2 – Understand
19	What is a Confusion Matrix? Explain its components.	CO4	L2 – Understand
20	Describe Hierarchical Clustering and its types.	CO4	L2 – Understand
21	What is Power BI and how is it used for Data Visualization?	CO5	L2 – Understand
22	Explain the process of getting data from different sources in Power BI.	CO5	L3 – Apply
23	Discuss various types of data visualizations in Power BI.	CO5	L2 – Understand
24	Describe the steps to publish and share reports in Power BI.	CO5	L3 – Apply

25	Give use cases of dashboards and analytical report creation using Power BI.	CO5	L3 – Apply
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